

- 1 When bananas ripen, they turn brown in colour. This is due to the formation of a brown product called melanin.

The enzyme catechol oxidase acts on its substrate, catechol, leading to the formation of a brown product, melanin, as shown in Fig. 1.1.

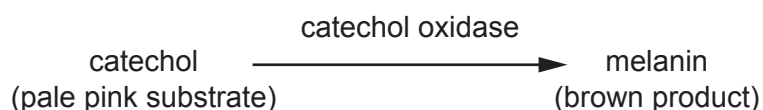


Fig. 1.1

As melanin is produced, the colour of the reaction mixture changes to brown. The intensity of the brown colour produced is proportional to the concentration of melanin. The more active the enzyme, the more intense the brown colour.

A colorimeter is used to measure the absorbance of the reaction mixture. Absorbance is a measure of the light absorbed by a coloured solution. With the reaction shown in Fig. 1.1, the more intense the brown colour, the higher the absorbance.

Catechol oxidase can be extracted from bananas.

- (a) Some students were asked to extract catechol oxidase by grinding a slice of banana with sand and water, using a mortar and pestle. The resulting mixture was then filtered through a piece of cloth.

Suggest a reason for:

grinding with sand

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filtering through cloth.

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..... [2]

- (b) The students investigated the effect of the substrate concentration on the enzyme-catalysed reaction shown in Fig. 1.1.

The students were provided with:

- the extracted catechol oxidase enzyme solution prepared in step (a), which was kept cold until needed
- a stock solution of 1.0% catechol solution made up in a buffered solution of pH 7.0
- a buffer solution of pH 7.0.

The procedure used by the students is outlined in steps 1 to 3.

1. The extracted catechol oxidase enzyme solution was mixed with the 1.0% catechol solution.
2. After 2 minutes a colorimeter was used to measure the absorbance of this mixture.
3. Step 1 and step 2 were repeated using different concentrations of catechol solution.

- (i) Suggest why the students used a colorimeter to measure the absorbance rather than judging the intensity of the colour by eye.

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..... [1]

- (ii) Identify the independent variable and the dependent variable in this investigation.

independent variable

dependent variable [2]

The students used the absorbance values at 2 minutes as the initial rates of reaction.

Fig. 1.2 shows the results.

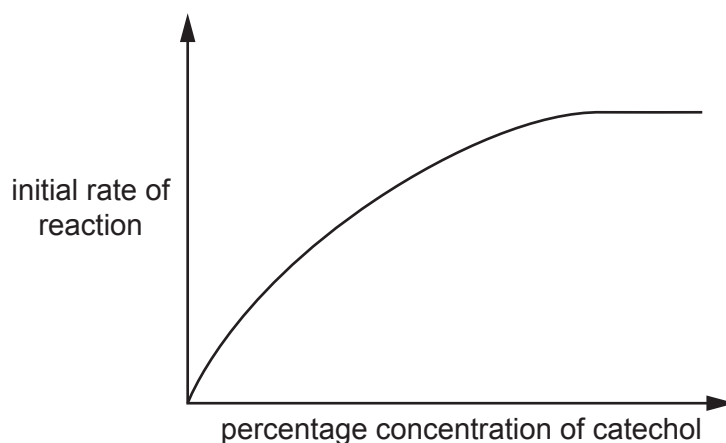


Fig. 1.2

- The description of your method should be set out in a logical way and be detailed enough for another person to follow.

You should **not** repeat the details from **(a)** describing how the extract of catechol oxidase is prepared.

This image shows a full page of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Fig. 1.3 shows the results when **no** inhibitor **Y** was added.

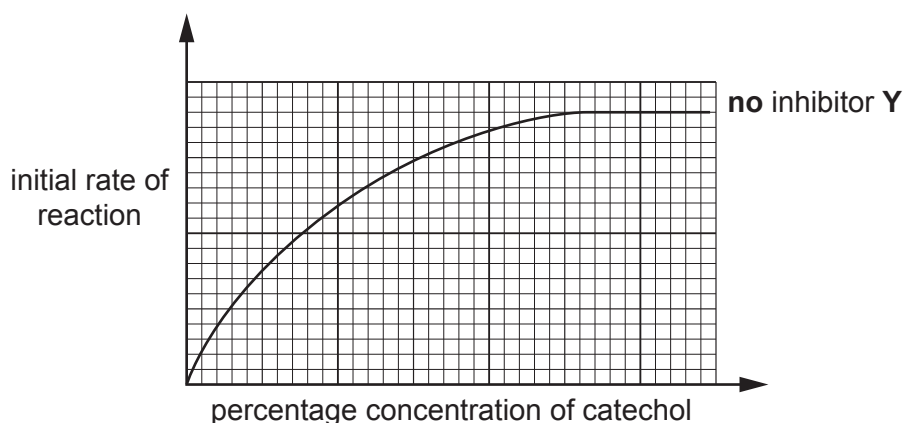


Fig. 1.3

The students suggested that inhibitor **Y** was acting as a **non-competitive** inhibitor.

- (ii) On Fig. 1.3, sketch the curve expected if inhibitor **Y** was acting as a non-competitive inhibitor. [2]

$V_{\max}$ is the maximum initial rate of reaction of the enzyme.

The Michaelis-Menten constant, K_m , is the substrate concentration at which the initial rate of reaction is half its maximum value.

- (iii) Draw on Fig. 1.3 the positions of $V_{\max}$ and K_m of the enzyme when **no** inhibitor **Y** is present. [2]
- (iv) Use your graph to describe the effect of the addition of inhibitor **Y** on the K_m of this enzyme.

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(d) Hypertension is a condition where people have long-term high blood pressure.

Two main types of drug are used to lower blood pressure in people with hypertension:

- ACE inhibitors
- diuretics.

Scientists investigated the effectiveness of the drugs when used separately **and** in combination.

505 people with hypertension were split into four groups as shown in Table 1.1.

Table 1.1

group	treatment
1	placebo (no active drug given)
2	ACE inhibitor only
3	diuretic only
4	ACE inhibitor and diuretic

The results are shown in Fig. 1.4.

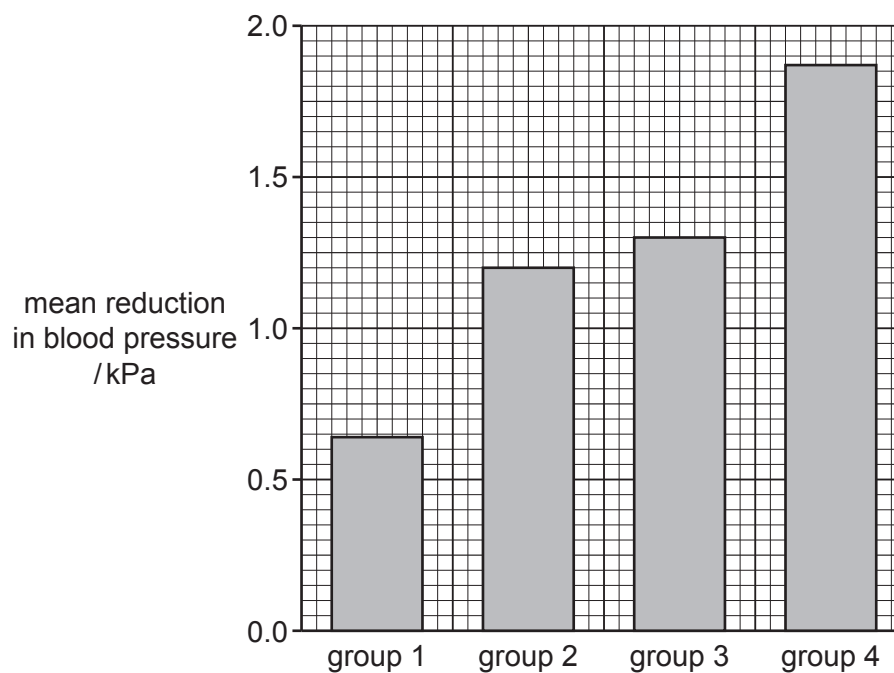


Fig. 1.4

The scientists analysed the data and concluded that a combination of an ACE inhibitor **and** a diuretic should be used to lower blood pressure in the treatment of hypertension.

With reference to the data in Table 1.1 and Fig. 1.4, discuss the conclusion that a combination of an ACE inhibitor and a diuretic should be used to lower blood pressure in the treatment of hypertension.

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[Total: 21]